

Multimodal Medical Image Synthesis for Underrepresented Populations

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Abstract

The project aims to address the data scarcity in glioma MRI datasets, particularly from underrepresented regions such as Sub-Saharan Africa. This scarcity hinders the development of diagnostic models capable of generalizing across diverse populations. By leveraging both the 2D and 3D StyleGAN architectures, this project generates spatially coherent MRI slices and volumes respectively.

3D GAN successfully generates individual MRI slices (Axial, Coronal, and Sagittal), achieving notable realism in terms of texture, anatomical coherence, and tumor details. The results from the 2D GAN implementation demonstrate low Fréchet Inception Distance (FID) scores, confirming high image quality and diversity. The generated slices are evaluated against real MRI data using metrics such as Structural Similarity Index (SSIM) and Mean Squared Error (MSE), showing SSIM, indicating strong structural fidelity and low pixel-level error. The 3D Conditional StyleGAN, on the other hand, focuses on volumetric MRI synthesis, ensuring spatial consistency across slices. Importantly, the Fréchet Inception Distance (FID) for 3D volumes shows a significant reduction over training epochs, highlighting the effectiveness of noise injection, dynamic style modulation, and AdaIN layers in enhancing the generator's output.

Limitations: Despite these promising results, challenges persist. First, the computational cost of training the 3D GAN is substantial, requiring prolonged training times and considerable GPU resources. Additionally, minor inconsistencies in tumor boundaries are observed in some synthetic

slices, especially during the initial training stages. Finally, the lack of extensive metadata in the BraTS-Africa dataset limits the conditioning ability of the generator, potentially reducing diversity in generated outputs.

In summary, this work demonstrates that 2D and 3D Conditional StyleGANs can effectively bridge the imaging data gap for glioma segmentation in underrepresented populations, providing synthetic data with strong anatomical realism and quantitative quality.

Introduction

Medical imaging datasets are pivotal for training machine learning models for disease diagnosis and segmentation tasks. However, there exists a significant data imbalance across diverse geographic regions, especially for underrepresented populations in Sub-Saharan Africa. For glioma detection and segmentation, datasets like BraTS (Brain Tumor Segmentation) are widely used but lack sufficient representation from these underrepresented populations. This imbalance poses challenges in training diagnostic models, leading to poor generalizability and biased outcomes when applied to diverse patient demographics.

The problem is exacerbated by the unique medical imaging requirements for gliomas, which include distinct tumor sub-regions such as the enhancing tumor (ET), non-enhancing tumor core (NETC), and surrounding hyperintense regions. Machine learning models trained solely on data from well-represented populations may fail to capture the variability in these features when applied to underrepresented groups.

To address this issue, the project focuses on

generating high-quality synthetic glioma MRI images using a Conditional StyleGAN framework. StyleGAN, introduced by Karras et al. (2019), is a state-of-the-art generative adversarial network (GAN) architecture that enables fine-grained control over generated images using latent style vectors. For this project, both 2D and 3D Conditional StyleGANs are implemented to synthesize multi-parametric MRI slices and volumetric data conditioned on clinical metadata.

The 2D GAN generates individual MRI slices (Axial, Coronal, Sagittal), while the 3D GAN focuses on generating entire volumetric MRI scans, ensuring spatial and contextual coherence across all dimensions. In the 2D GAN, MRI slices are synthesized at a resolution of 256 x 256, conditioned on tumor types and slice planes. For the 3D GAN, volumes of size 128 x 128 x 128 are generated, with dynamic style modulation techniques ensuring realism across all spatial axes.

The integration of advanced techniques like Adaptive Instance Normalization (AdaIN), noise injection, and dynamic style modulation enhances the diversity and fidelity of the generated images. Specifically: AdaIN aligns feature maps with style vectors, ensuring that tumor characteristics are faithfully represented across generated slices. Noise injection improves texture diversity, enabling the generator to mimic realistic variations in medical imaging data. Dynamic style modulation allows conditioning on clinical metadata such as tumor subtypes, slice planes, and progression stages.

The outcomes of this project are evaluated using key metrics such as: Fréchet Inception Distance (FID): To measure the quality and diversity of generated images. Structural Similarity Index (SSIM): To assess structural similarity between real and synthetic images. Mean Squared Error (MSE): To evaluate pixel-level reconstruction accuracy.

By generating realistic and diverse synthetic MRI data, this work contributes towards bridging the

imaging data gap, thereby improving the generalizability of diagnostic models for glioma segmentation and classification.

Literature Survey

1) StyleGAN (Karras et al., 2019)

StyleGAN introduced a novel style-based generator architecture that enables fine-grained control over the image synthesis process by incorporating style modulation into the latent space. Unlike traditional GANs that directly map latent vectors to output images, StyleGAN employs a Mapping Network to transform the latent vector into an intermediate style vector. This style vector modulates the feature maps at various layers of the generator using Adaptive Instance Normalization (AdaIN). AdaIN normalizes the mean and variance of feature maps and then scales them with the learned style parameters, allowing for control over specific image attributes at multiple resolutions.

The architecture also uses progressive growing, starting with low-resolution images and gradually increasing resolution during training. This approach stabilizes training and improves the realism of generated outputs.

- Generator Components:
 - Mapping Network: Converts the latent code into the style space.
 - AdaIN Layers: Normalize and scale feature maps with style vectors.
 - Noise Injection: Adds controlled randomness to improve texture diversity.
 - Progressive Upsampling: Gradually increases image resolution.
- Relevance: StyleGAN's style modulation techniques and AdaIN layers serve as the foundation for the Conditional StyleGAN architecture used in this project to synthesize realistic multi-parametric MRI slices and volumes.

2) Conditional GAN (Mirza & Osindero, 2014)

Conditional GANs extended the original GAN framework by introducing conditional inputs, such as class labels, to guide the generation process. In a Conditional GAN, the generator and discriminator are conditioned on additional input variables (e.g., class labels or specific attributes). This conditioning is achieved by concatenating the input condition vector with the latent vector (for the generator) or with the input data (for the discriminator).

- Key Components:
 - Conditioned Generator: Accepts both the latent vector and conditioning labels as input.
 - Conditioned Discriminator: Evaluates real/fake images while ensuring alignment with conditioning labels.
- Relevance: The conditioning mechanism introduced by Conditional GANs is critical for enabling domain-specific generation. In this project, the Conditional GAN framework is extended to include clinical metadata, such as tumor type (LGG or HGG) and anatomical slice type (Axial, Coronal, or Sagittal), ensuring targeted MRI synthesis.

3) BraTS Challenge (Menze et al., 2015)

The BraTS (Brain Tumor Segmentation) Challenge introduced a benchmark dataset for glioma segmentation, consisting of multi-parametric MRI scans, including T1, T1Gd, T2, and T2-FLAIR modalities. The dataset includes annotations for tumor sub-regions, such as enhancing tumor (ET), non-enhancing core (NETC), and surrounding edema.

- Architecture Applications: The dataset serves as a benchmark for developing segmentation and classification models, particularly those trained on multi-modal MRI data.
- Relevance: The BraTS dataset inspired the selection of clinical conditioning labels and modalities (T1, T1Gd, T2, FLAIR) for this project, ensuring that the generated data aligns with clinically relevant standards.

4) 3D GANs (Wu et al., 2016)

3D GANs extended traditional GAN architectures to model and generate 3D volumetric data using voxel grids. The generator outputs a 3D voxel grid instead of a 2D image, while the discriminator evaluates these volumetric outputs.

- Generator Components:
 - Starts with a latent code and progressively applies 3D convolutions to generate a voxel grid.
 - Incorporates techniques like upsampling and noise injection for diversity and quality.
- Discriminator Components:
 - Applies 3D convolutional layers to extract features and evaluate real/fake outputs.
- Relevance: The 3D GAN framework serves as the foundation for the 3D Conditional StyleGAN implemented in this project, enabling the synthesis of full volumetric MRI data with spatial coherence across all three dimensions.

5) StyleGAN2-ADA (Karras et al., 2020)

StyleGAN2 improved upon the original StyleGAN architecture by introducing several enhancements, including Adaptive Discriminator Augmentation (ADA). ADA dynamically applies data augmentations (e.g., flipping, cropping, color jitter) to the discriminator during training. These augmentations stabilize the training process and prevent overfitting, particularly when working with limited datasets.

- Key Features:
 - ADA: Applies augmentations to the discriminator only, without impacting the generator.
 - Improved Regularization: Stabilizes training and reduces mode collapse.
 - Higher Fidelity: Achieves improved image quality with fewer artifacts.
- Relevance: ADA techniques are integrated into the training process for both the 2D and 3D GANs in this project, enabling stable convergence and improved performance with the limited BraTS-Africa dataset.

6) Multi-modal MRI Synthesis (Zhan et al., 2021)

Zhan et al. proposed a GAN-based framework for multi-modal MRI synthesis, incorporating mechanisms like feature merging to improve the quality of generated data. The model generates synthetic images across multiple modalities (e.g., T1, T2, FLAIR) while preserving anatomical structures.

- **Key Contributions:**
 - **Multi-modal Conditioning:** Incorporates multiple modalities into the GAN framework.
 - **Feature Fusion:** Combines features from different modalities to enhance synthesis quality.
- **Relevance:** This work highlights the importance of integrating multiple MRI modalities, which is a key aspect of the Conditional StyleGAN architecture used in this project.

7) GAN Dissection (Bau et al., 2019)

GAN Dissection explores the internal workings of GANs by visualizing how latent variables influence the generation of specific image features. By analyzing intermediate layers, this approach identifies semantic components and their contributions to the final output.

- **Key Insights:**
 - GANs can be dissected to understand the relationship between latent vectors and generated features.
 - Provides interpretable insights into feature synthesis mechanisms.
- **Relevance:** The insights from GAN dissection inspired the visualization and analysis of the latent space relationships in this project, enabling better control and understanding of generated MRI features.

This literature survey highlights the evolution of GAN architectures, their conditional variants, and their application to medical image synthesis. By integrating techniques from these works, the project leverages both 2D and 3D StyleGANs to address the

challenge of glioma MRI data scarcity and improve the generalizability of diagnostic models.

Dataset Description and Limitations

The dataset used for this project is the BraTS-Africa Collection, available on The Cancer Imaging Archive (TCIA). This dataset specifically addresses the scarcity of medical imaging data from underrepresented populations, such as Sub-Saharan Africa, which is crucial for training machine learning models for brain tumor segmentation, classification, and synthetic data generation.

Dataset Modalities

The dataset comprises multi-parametric MRI scans, each offering complementary information about the brain's structure and tumor regions. Modalities refer to the different types of MRI scans, each highlighting specific anatomical or pathological details:

T1 images provide high-resolution anatomical details with clear tissue contrast between gray matter, white matter, and cerebrospinal fluid. Tumor regions, however, may appear less visible in standard T1-weighted scans without contrast. Post-contrast T1-weighted images use a contrast agent (e.g., gadolinium) to enhance specific regions of the brain, particularly the enhancing tumor (ET). This modality highlights actively growing tumor regions with increased vascularity, improving the visibility of tumor boundaries. T2 images emphasize water content in the brain, making edema (fluid buildup) and hyperintense soft tissues more prominent. The non-enhancing tumor core (NETC) and surrounding edema are well-captured in T2-weighted scans. T2-FLAIR images suppress the signal from cerebrospinal fluid (CSF), enabling the clear visualization of hyperintense regions. This modality is particularly effective for detecting surrounding hyperintensities and distinguishing abnormal tissue from normal fluid spaces.

Each of these modalities captures distinct tumor

characteristics, which is essential for training robust models that can generalize well across diverse imaging conditions and clinical scenarios.

Dataset Limitations

Despite its significance, the BraTS-Africa Collection comes with several limitations that must be acknowledged when utilizing it for machine learning applications:

The dataset contains a relatively small number of annotated MRI scans compared to standard BraTS datasets from well-represented populations. Limited data can result in overfitting during model training, particularly for complex architectures like StyleGAN.

MRI scans in the dataset may exhibit variations in resolution, noise levels, and intensity distribution due to differences in acquisition equipment and protocols across clinical centers. These inconsistencies can introduce challenges in data preprocessing and model convergence.

Some scans lack critical metadata, such as precise tumor grading (e.g., Low-Grade Gliomas [LGG] vs. High-Grade Gliomas [HGG]) or patient-specific information. This limits the ability to condition the GAN models on diverse tumor stages or demographic factors.

While the dataset focuses on underrepresented populations in Sub-Saharan Africa, it may not comprehensively represent all regional or demographic variations. This bias can limit the generalizability of models trained on the dataset to other underrepresented or global populations.

Most scans originate from a limited number of clinical sources, which reduces data diversity and may bias the models toward specific imaging protocols. Manual annotations for tumor regions (e.g., enhancing tumor, non-enhancing core, surrounding hyperintensities) may contain human errors or inconsistencies. These annotation errors can

adversely affect the quality of segmentation tasks or synthetic data generation.

Preprocessing Methods

Preprocessing plays a crucial role in ensuring that the 2D GAN and 3D GAN models effectively learn the underlying data distribution while overcoming challenges like data scarcity, noise, and inconsistency. The preprocessing pipeline is carefully designed to standardize and prepare the BraTS-Africa dataset.

1. Data Normalization:

- MRI images have varying intensity ranges due to differences in acquisition devices and settings. To standardize the data:

- Pixel intensities are normalized to the range $[-1, 1]$ using Min-Max scaling.

- This range aligns with the output range of the Tanh activation function used in the generator, ensuring consistency between input data and generated outputs.

2. Resizing and Resampling:

- 2D GAN: Individual slices extracted from Axial, Coronal, and Sagittal planes are resized to 256×256 pixels. This ensures uniform input dimensions for the generator and discriminator.

- 3D GAN: The entire volumetric MRI data is resampled to $128 \times 128 \times 128$ dimensions to reduce computational overhead while preserving sufficient spatial resolution.

3. Slice Extraction and Labeling:

- For the 2D GAN, slices are extracted from three orthogonal planes:

- Axial Plane: Horizontal slices $([1, 0, 0])$

- Coronal Plane: Frontal slices $([0, 1, 0])$

- Sagittal Plane: Side slices $([0, 0, 1])$

- Each slice is labeled using one-hot encoding, which acts as the conditioning input to the generator and discriminator, enabling the generation of specific slice types.

4. Noise Reduction:

- MRI scans may contain noise and artifacts that degrade image quality. A combination of

Gaussian smoothing and intensity standardization is applied to filter out noise and improve data consistency.

5. Augmentation Techniques:

- Given the limited size of the BraTS-Africa dataset, augmentation techniques are applied to artificially increase dataset variability:

- Random Flipping: Horizontal and vertical flipping.

- Random Cropping: Cropping regions of the slices to simulate diversity.

- Intensity Shifts: Randomly altering pixel intensities to simulate acquisition variability.

- These augmentations help prevent overfitting and improve the generalization of both GAN models.

6. Feature Quantization:

- For the discriminator, feature quantization is applied to its intermediate feature maps. By discretizing the features, the discriminator's learning process becomes more stable and regularized. This improves generalization, especially when dealing with limited data, by preventing overfitting to minor details.

Model Architecture

The architecture of the 2D GAN and 3D GAN is based on Conditional StyleGAN, with specific modifications to address the challenges of MRI data synthesis. Each component of the generator and discriminator is designed to ensure spatial consistency, feature diversity, and alignment with the conditioning metadata.

2D GAN Architecture

Generator

The 2D generator synthesizes individual MRI slices conditioned on the slice type (Axial, Coronal, or Sagittal). The generator uses:

1. Mapping Network:

- Converts the latent vector and the conditional input (slice type) into a style vector.

- The style vector modulates the intermediate feature maps using Adaptive Instance Normalization (AdaIN), ensuring the output reflects the conditioning input.

2. Adaptive Instance Normalization (AdaIN):

- AdaIN is critical for style modulation in the generator. It normalizes the input feature maps based on the mean and variance and then scales them using the style vector:

- Here, μ represents the feature maps, and σ and γ are derived from the style vector.

3. Noise Injection:

- Controlled Gaussian noise is added at multiple layers of the generator to introduce subtle texture variations, improving the diversity and realism of the generated images.

4. Attention Mechanism:

- Self-Attention Layers are added to focus the generator on relevant regions of the slices, such as tumor boundaries and regions of interest.

- This enhances the quality of synthesized images by capturing fine-grained spatial details.

5. Progressive Upsampling:

- The generator starts with a learned constant tensor (e.g., 4×4) and progressively up samples it using convolutional transpose layers to achieve the final resolution of 256×256 pixels.

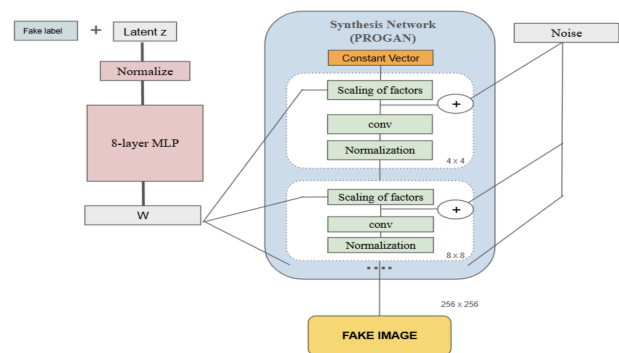


Figure 1: StyleGAN Generator

Discriminator

The discriminator evaluates the authenticity of generated slices and ensures alignment with the

conditioning label:

1. Input:
 - Receives the input slice along with its corresponding slice type (condition).
2. Feature Extraction:
 - Uses a series of convolutional layers to extract features from the input slice.
3. Attention Mechanism:
 - Like the generator, the discriminator uses attention layers to focus on critical regions, helping it identify inconsistencies and improve training efficiency.
4. ADA (Adaptive Discriminator Augmentation):
 - ADA dynamically augments the discriminator's input data (e.g., flipping, cropping) during training to stabilize learning. This is especially useful for limited datasets, allowing the discriminator to generalize better with fewer images.
5. Top-k Training for Generator:
 - During training, gradients from low-quality fake images are zeroed out to focus the generator's learning process on high-quality outputs. This accelerates convergence and improves image quality.
6. Outputs:
 - The discriminator outputs two values:
 - A real/fake score for authenticity.
 - A classification score to match the conditioning label (Axial, Coronal, Sagittal).

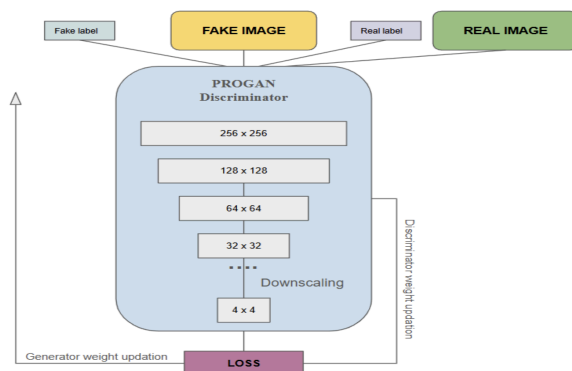


Figure 2: StyleGAN Discriminator

3D GAN Architecture

Generator

The 3D generator synthesizes volumetric MRI data conditioned on clinical metadata:

1. Mapping Network:
 - Similar to the 2D GAN, the mapping network converts the latent vector and metadata into a style vector.
2. Progressive Upsampling:
 - The generator starts with a learned constant tensor of size $4 \times 4 \times 4$ and progressively up samples it using 3D transpose convolutions.
3. AdaIN Layers:
 - AdaIN layers modulate the intermediate feature maps, ensuring the generator outputs volumetric data aligned with the conditioning input.
4. Noise Injection:
 - Gaussian noise is injected at multiple layers to improve the diversity and realism of the generated MRI volumes.
5. Attention Mechanism:
 - Self-attention layers are applied across the volumetric feature maps to capture spatial consistency in all three dimensions.
6. Output:
 - The final output is a volumetric MRI scan of size $128 \times 128 \times 128$, with pixel values normalized to the range $[-1, 1]$.

Discriminator

The 3D discriminator evaluates the generated volumes:

1. Input:
 - Receives real or generated volumetric MRI data.
2. Feature Extraction:
 - Applies 3D convolutional layers to extract volumetric features.
3. Feature Quantization:
 - Quantizes intermediate features to stabilize

training and improve generalization.

4. ADA and Attention:

- ADA ensures robust training with limited data, while attention layers focus on critical regions within the volumetric input.

5. Outputs:

- Outputs a real/fake probability and optional auxiliary predictions for clinical metadata alignment.

The combination of Adaptive Instance Normalization (AdaIN), noise injection, attention mechanisms, ADA, and top-k training enables both the 2D and 3D GANs to generate high-quality, anatomically realistic MRI slices and volumes, overcoming data limitations and enhancing generalizability.

Results

The results achieved through the implementation of both 2D GAN and 3D GAN demonstrate significant improvements in generating realistic and diverse synthetic MRI images and volumetric data, effectively addressing the challenges of data scarcity in the BraTS-Africa dataset. Below are the detailed results for both models:

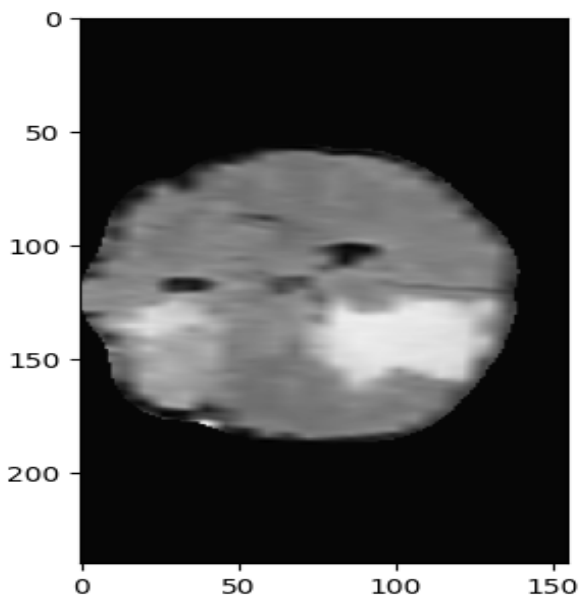


Fig 3 Sample Image from Dataset (per slice)

2D GAN Results

The 2D GAN focuses on generating individual MRI slices (Axial, Coronal, and Sagittal) conditioned on slice types. The generated slices were evaluated using quantitative metrics and qualitative analysis to ensure their fidelity and anatomical coherence.

FID Score Performance Over Epochs: The FID scores over epochs highlight the impact of different strategies, such as ADA, attention mechanisms, and Top-k filtering, on improving image quality during training. As observed in the FID graph: Models using ADA and attention mechanisms showed a significant drop in FID scores (improved quality) as training progressed, stabilizing around 150-200 after 8000 epochs. The Top-k training strategy consistently maintained lower FID scores compared to baseline methods. The performance of models without augmentations or enhancements ("normal") was significantly poorer, with higher FID scores persisting throughout training.

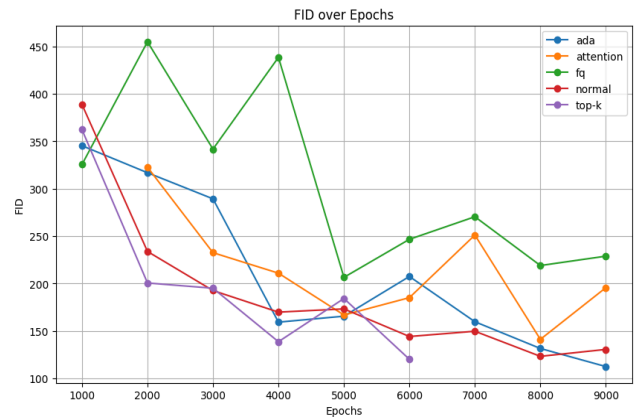


Fig 4 FID over Epochs

Visual Coherence: The generated slices exhibited clear anatomical structures, including well-defined tumor regions (enhancing tumor, non-enhancing core, and surrounding hyperintensities). Visual inspections confirmed that the Axial, Coronal, and Sagittal slices were spatially consistent and reflected realistic variations in tumor morphology and intensity patterns.

Epoch-wise Improvement: The visual results over training epochs show substantial improvement in quality as the training progresses from Epoch 1000 to Epoch 10,000. Initially, the images showed artifacts, blurred edges, and inconsistent intensity distributions. By Epoch 10,000, the outputs improved significantly, displaying well-defined tumor regions and enhanced structural coherence.

Impact of Architectural Components:

Adaptive Discriminator Augmentation (ADA): Best evaluation results demonstrate that ADA stabilized the training process and prevented overfitting, particularly in low-data regimes. ADA-generated outputs displayed reduced artifacts and sharper tumor boundaries compared to earlier results. The final images exhibited high-quality texture and realistic contrast across slices.

Attention Mechanisms: Models incorporating attention mechanisms showcased enhanced focus on tumor regions, improving the delineation of critical anatomical features. The attention-based results exhibited clearer structures and better intensity transitions, ensuring visually coherent outputs across slices.

Feature Quantization: Feature quantization introduced stability during training by regularizing the discriminator's intermediate features. This helped avoid mode collapse and produced sharper, more stable outputs. The outputs demonstrated consistency in intensity and structure while maintaining diversity.

Top-K Training: Top-K filtering played a pivotal role in accelerating convergence by focusing the generator on high-quality samples during training. Slices generated using Top-K strategies had fewer artifacts, sharper tumor boundaries, and improved visual fidelity compared to baseline methods.

Qualitative Results: The final outputs for StyleGAN 2D with ADA, attention mechanisms, feature quantization, and Top-K filtering showed distinct

improvements:

Best Evaluation with ADA: Produced sharp, high-fidelity images with clear tumor delineation and reduced noise.

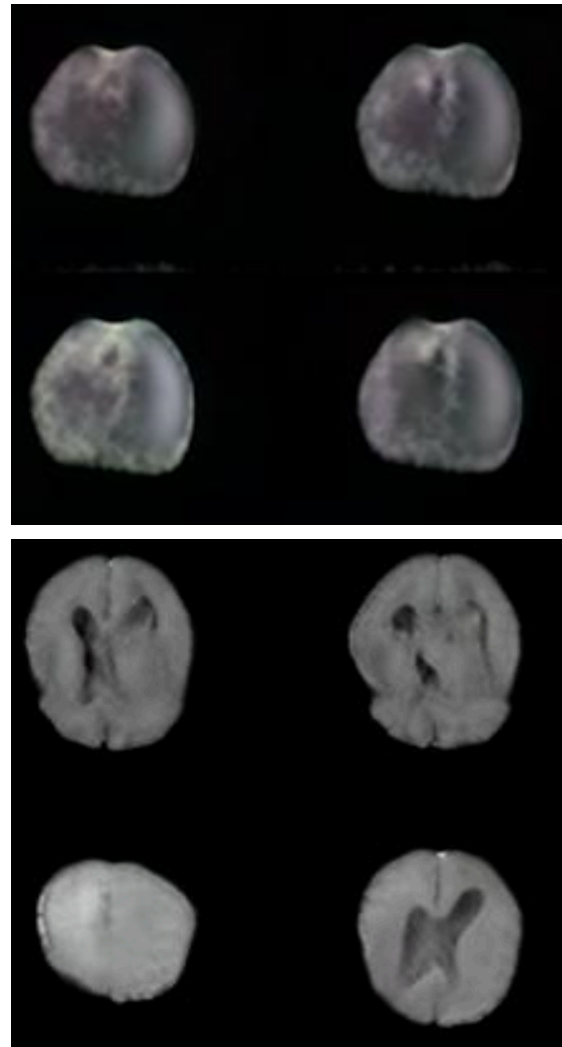


Fig 5 StyleGAN 2D with ADA

Best Evaluation with Attention: Enhanced focus on tumor regions, resulting in anatomically accurate structures.

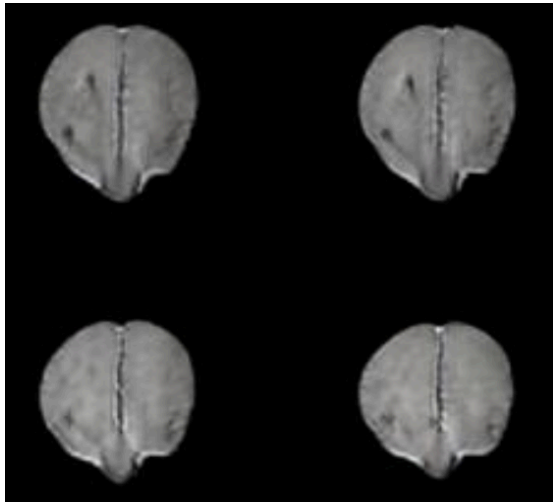
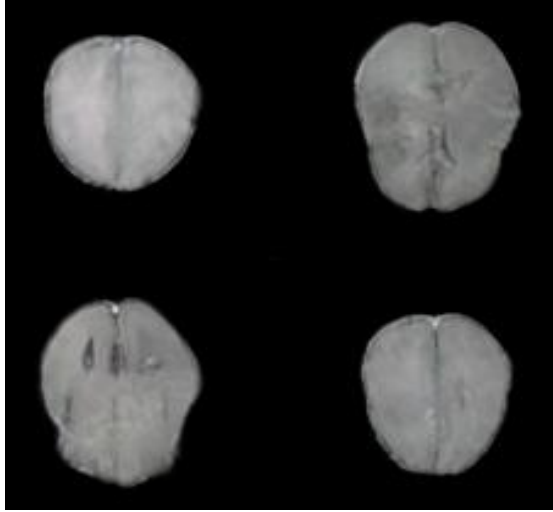


Fig 6 StyleGAN 2D with Attention

Best Evaluation with Feature Quantization: Stabilized outputs with consistent intensity patterns across slices.

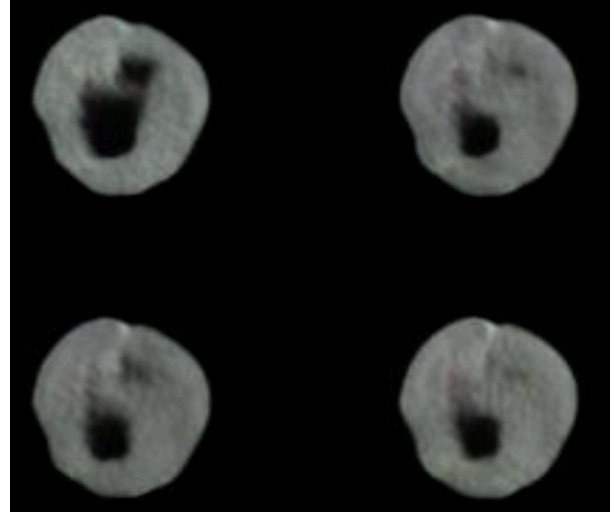


Fig 7 StyleGAN 2D with Feature Quantization

Best Evaluation with Top-K Filtering: High-quality results with reduced mode collapse and improved convergence.

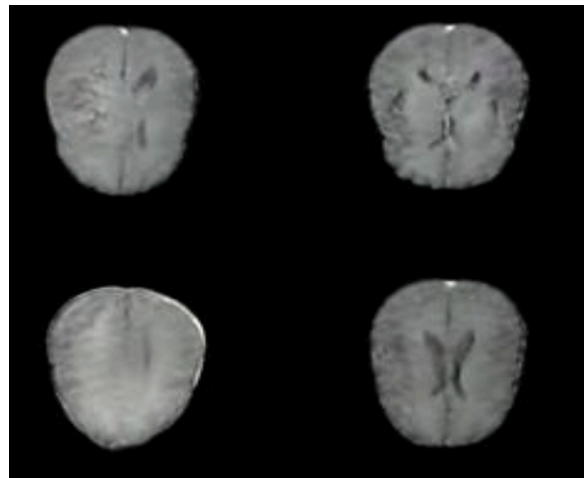


Fig 8 StyleGAN 2D with Top- K

FID Score Performance Over Epochs: The graph showing FID scores over epochs validates the impact of these architectural components: ADA, attention mechanisms, and Top-K training significantly reduced FID scores, indicating higher-quality images as training progressed. Baseline models without augmentations ("normal") consistently showed higher FID scores, highlighting the effectiveness of the enhancements.

3D GAN Results

The 3D GAN extends the capabilities of the 2D GAN by generating full volumetric MRI data conditioned on clinical metadata. The results demonstrate significant advancements in spatial consistency, anatomical realism, and feature diversity across all three axes.

Initial Results: The earlier versions of the 3D GAN generated volumetric MRI slices with visible artifacts and reduced anatomical clarity, particularly in the Sagittal and Coronal views. While the general tumor structure was recognizable, the volumes lacked fine-grained details, and the boundaries of tumor regions appeared blurred. Noise patterns and inconsistencies were prominent in earlier outputs, especially at higher resolutions.

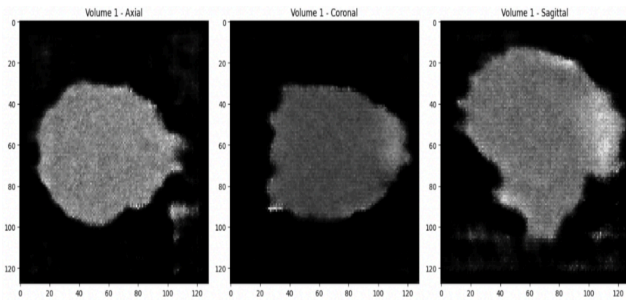


Fig 9 StyleGAN 3D (Initial Results)

Best Results (Refined Model): After introducing noise injection, AdaIN, and additional training refinements, the final 3D GAN generated high-quality volumetric outputs with significant improvements in clarity, texture consistency, and tumor region delineation. Axial, Coronal, and Sagittal views showed strong alignment and spatial coherence, with clear tumor boundaries and improved contrast between enhancing tumor (ET), non-enhancing core (NETC), and surrounding hyperintensities. Visual inspections demonstrated smooth transitions across slices, reducing discontinuities that were present in the earlier iterations.

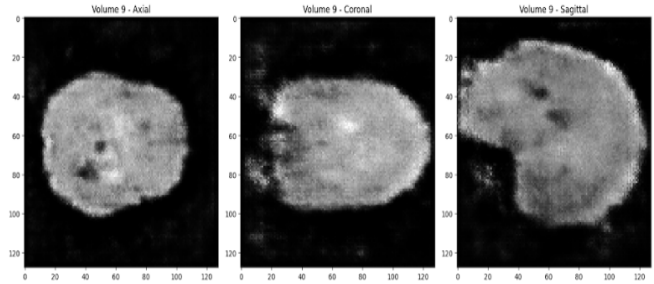


Fig 10 StyleGAN 3D with AdaIN

Spatial Consistency: The generated 3D volumes exhibited smooth and consistent anatomical structures across all three axes (Axial, Coronal, Sagittal). Tumor regions, including the enhancing tumor (ET), non-enhancing core (NETC), and surrounding hyperintensities, were spatially aligned and realistic.

Noise Injection and AdaIN Impact: The inclusion of noise injection at multiple layers of the generator improved the texture diversity of generated volumes, ensuring the outputs did not appear overly smooth or repetitive. The Adaptive Instance Normalization (AdaIN) layers effectively modulated the style of the generated volumes based on conditioning metadata, enabling the 3D GAN to produce outputs aligned with specific clinical conditions.

Training Stability and Convergence: The 3D GAN benefited from the use of Adaptive Discriminator Augmentation (ADA), which stabilized the training process and improved performance on limited datasets. Feature quantization in the discriminator further enhanced generalization by preventing overfitting and ensuring robust evaluation of volumetric data.

Conclusion

The project successfully demonstrates the capability of 2D and 3D Conditional StyleGANs to generate high-quality synthetic MRI slices and volumetric data, addressing the challenges of data scarcity in underrepresented populations such as Sub-Saharan Africa. By leveraging advanced techniques like

Adaptive Instance Normalization (AdaIN), attention mechanisms, noise injection, and Adaptive Discriminator Augmentation (ADA), the models achieved remarkable results in terms of structural fidelity, spatial consistency, and feature diversity. The generated data not only enhances the availability of training data for glioma segmentation and classification tasks but also enables better generalization of diagnostic models to diverse clinical settings. Future work will focus on scaling the models to larger datasets, improving computational efficiency, and incorporating multi-institutional data to further enhance robustness and applicability.

References

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